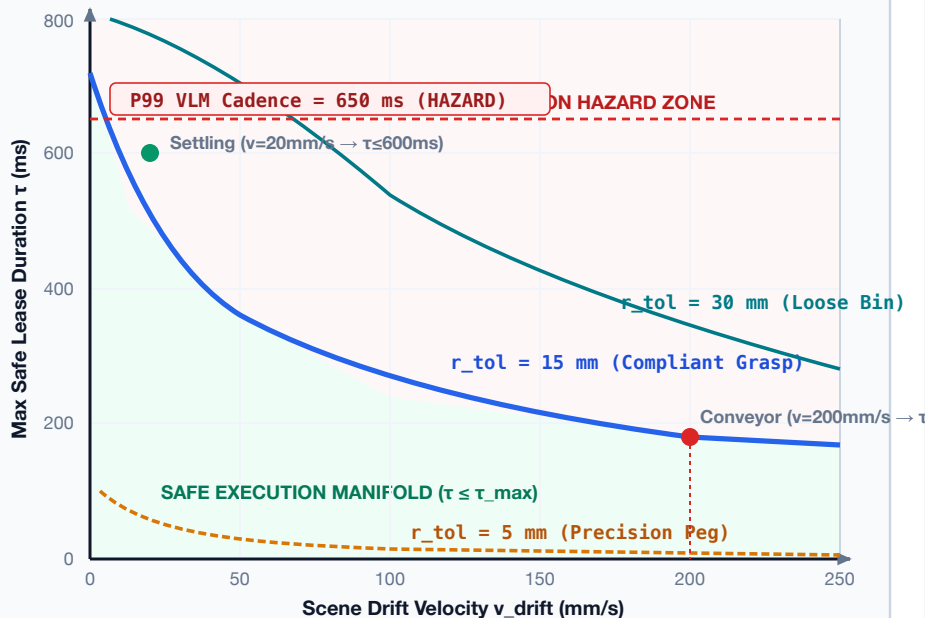


INTENT LEASE DYNAMICS & EARLY ADMISSIBILITY FILTERING

Physics-Derived TTL Horizon vs. O(1) Kinematic & Dynamic Ingestion Gateways

SCENE PHYSICS · DYNAMICS

PANEL A: LEASE DURATION vs. DRIFT



Mathematical Derivation: Maximum Safe TTL (§10.4)

$$\tau_{\text{max}} = (r_{\text{tol}} - \sigma_{\text{sensor}}) / v_{\text{drift}}$$

Design Rule: Lease validity is an invariant of **scene physics**, not model compute speed.
If inference latency jitters ($P99 > \tau_{\text{max}}$), system must halt smoothly rather than stretch lease.

INGESTION GATE · ARCHITECTURE

PANEL B: EARLY REJECTION FILTER

SYSTEM 2 VLM / BRAIN

Intent Packet ($\text{seq}, t_{\text{src}}, \text{TESE}(3), \Sigma, \tau, F_{\text{max}}$)

168 Bytes

< 1 μs

1. Monotonicity & Hash Gate

$\text{seq} > \text{seq}_{\text{curr}} \wedge \text{parent}_{\text{hash}} == \text{ring}_{\text{buffer}}[t_{\text{src}}]$

Drop Stale/Replay

O(1) · 5 μs

2. Kinematic Workspace & Occupancy

$p_{\text{target}} \in \mathcal{W}_{\text{reach}} \wedge \text{Occupancy}(p_{\text{target}}) == \text{VALID}$

Out-of-Reach Reject

< 2 μs

3. Dynamic Reachability Bound

$t_{\text{min}} = 2\sqrt{(d/a_{\text{max}})} \leq \tau_{\text{rem}}$ (Under thermal limits)

Time Insufficient

< 1 μs

4. Covariance & Ambiguity Gate

$\lambda_{\text{max}}(\Sigma) \leq \sigma^2_{\text{max}} \wedge \text{Softmax Entropy } H(Y) \leq H_{\text{thresh}}$

Refusal on Distractor

✓ ADMITTED INTENT RECORD

- Dispatched to 50 Hz Trajectory Optimizer
 - Enforced by 1000 Hz Real-Time MCU Shield
- Total Ingestion Overhead: 42 μs (P99)**

✗ STRUCTURED REJECTION

- Diagnostic Feedback to System 2 Reasoner
 - Returns Shortfall ($t_{\text{min}} - \tau$) & Margin
- Prevents Solver Queue Starvation**

Computational Cost Asymmetry (§10.5)

Early Rejection Gate (Ingestion): 42 μs scalar processor lookup. Zero queue bubble.

Late Rejection (Planner Interior): 15–80 ms solver penalty search. Starves real-time pipeline.

Ingestion filtering guarantees downstream solver only executes kinematically solvable goals.